Linda Mansour

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Education

University of Illinois at Chicago Graduated May 2025
Bachelor of Science in Data Science - Concentration in Bioinformatics

Technical Skills

Programming Languages: Python, R, C/C++, MySQL, Matlab, SAS, HTML/CSS
Bioinformatics Tools: BLAST, Semblans, RNA-Seq Analysis, Shell Scripting
Machine Learning Frameworks: Pandas, Numpy, Matplotlib, Scikit-learn, TensorFlow, Seaborn, Gurobi
General Skills: Data Processing and Visualization, Version Control with Git/GitHub, Linux/Unix, API Integration, Workflow Automation, Data Cleaning, Markdown

Research Experience

Computational Biology Lab Research Assistant September 2023 – Present
University of Illinois at Chicago *Chicago, IL*

- Developing Python-based methods to process high-throughput sequencing data to better understand data availability within public genetic databases.
- Facilitate weekly meetings with lab members and the advisor to apply methods and build and revise pipelines to streamline lab processes involving large amounts of genomic and transcriptomic data.
- Implement BLAST, Salmon, and Kraken pipelines to classify sequences and contribute to scientific papers.
- Processed over 200 million sequences using automated workflows for classification and alignment tasks.

Gene Expression Research Project January 2024 – June 2024
University of Illinois at Chicago *Chicago, IL*

- Investigated differential gene expression across plant lineages, applying bioinformatics pipelines for transcriptomic analysis.
- Presented findings as a poster at Botany 2024, engaging with experts in computational biology and plant genomics.
- Performed principal component analysis (PCA) and generated expression heatmaps to visualize sample clustering and dominant transcriptional patterns.

Publications

Herbaria provide a valuable resource for obtaining informative mRNA February 2025
Alexa S. Tyszka, Khong-Sam Chia, Eric C. Bretz, Linda Mansour, Drew A. Larson, Philip Carella, Joseph F. Walker
bioRxiv 2025.02.12.637878; doi: 10.1101/2025.02.12.637878

Teaching Experience

Populations and Communities Biology Teaching Assistant August 2025 – Present
University of Illinois at Biological Sciences Department *Chicago, IL*

- Attend weekly meetings with supervising professor to revise and improve classroom learning materials and presentations.
- Grade assignments, quizzes, and lab reports in addition to providing constructive feedback on student work.
- Host office hours several times a week to further support students in understanding lecture or lab materials.

Undergraduate Teaching Assistant September 2022 – May 2023
University of Illinois at Chicago Computer Science Department *Chicago, IL*

- Mentored 460 students with Python fundamentals, labs, and coding projects, hosting office hours for 15+ students weekly.
- Instructed labs weekly and delivered insightful feedback to students, teaching assistants, and professors to foster academic success and a collaborative environment.
- Evaluated and graded over 1,000 assignments, labs, and exams while providing weekly one-on-one sessions to help struggling students, improving comprehension and student success.

Projects

MissMap March 2025 – Ongoing
Phylogenetic Data Pipeline in Python

- Developing a Python pipeline to retrieve and summarize sequence availability in a visually accessible matrix from NCBI GenBank across multiple plant clades, enabling large scale phylogenetic data assessment.

United States Wildfire Analysis August 2024 – December 2024
Machine Learning and Data Visualization in Python

- Processed 900,000 wildfire records, integrating 11,000 weather data points via 40,000 Open-Meteo API calls for risk analysis.
- Engineered 15 predictive features and trained random forest models to analyze environmental correlations.

Personal Portfolio - lindascomputing.xyz December 2023 – Ongoing
React and HTML/CSS Development

- Designed a React-based portfolio that showcases past coding projects, Rosalind solutions, and personal information.

Grants and Fellowships

Walker Excellence in Computational Biology Award 2024 & 2025

University of Illinois at Chicago

- *Awarded $1,500 for outstanding contributions to computational biology research. Awarded in 2024 and 2025.*

National Science Foundation Graduate Research Fellowship Program (NSF GRFP) *October 2024*

Proposal Submission

- *Developed and submitted an independent research proposal on synteny-based phylogenetics.*

Conference Contributions

Botany 2024 June 2024

Grand Rapids, MI

- Poster Presentation: Examining gene expression patterns in plant tissues after storage in silica gel.

Botany 2025 July 2025

Palm Springs, CA

- Poster Presentation: MissMap: A pipeline for visualizing data availability in plant clades.